Week 1: Data Cleaning and Feature Engineering Report

Group-9 Intern's Names :

Muhammad Subhan Khan

Sasikanth Danthanala

Olawale Owoyomi

Omotosho Johnson

Swati Mani

Sania Farheen

Helen Akpabio Ndarake

Muhammad Tafoor

Excelerate – AI Powered Data Analysis Internship

Date of Submission: 19/01/2026

1. Introduction:

The purpose of this report is to document the data cleaning and feature engineering activities completed during Week 1 of the internship. The primary objective of this week was to transform the raw dataset into a clean, structured, and analysis-ready format that can be used for further exploratory and analytical tasks in upcoming weeks.

The dataset used for this task is titled **"SLU Opportunity Wise Data"**. It contains learner-level information related to various opportunities, including signup details, demographic attributes, application dates, and opportunity timelines. The dataset consists of approximately 8,599 records and 16 columns, covering both categorical and date-based attributes.

2. Data Description:

The dataset includes the following key types of information:

2.1. Learner Information

• First Name: Identifies the learner and is mainly used for record reference.

• Date of Birth: Used to calculate learner age for demographic analysis.

• Gender: Represents the gender of the learner.

• Country: Indicates the learner's country of residence.

• Institution Name: Shows the educational institution the learner is associated with.

• current/Intended Major: Describes the learner's field of study or academic interest.

2.2. Opportunity Details

• Opportunity Id: A unique identifier for each opportunity.

• Opportunity Name: The name of the course or program applied for.

• Opportunity Category: Specifies the type of opportunity, such as a course.

• Opportunity Start Date: The date when the opportunity begins.

• Opportunity End Date: The date when the opportunity ends.

These fields help group and analyze learners across different opportunities.

2.3. Application Timeline

• Learner Sign-up Date Time: The date and time when the learner registered on the

platform.

• Apply Date: The date when the learner submitted the application.

• Entry created at: The date when the application record was created.

These columns are useful for analyzing application trends and learner engagement over time.

2.4. Application Status

• Status Description: Describes the learner's current stage in the opportunity.

• Status Code: A numeric value representing the application status.

These fields help track learner progress and identify outcomes such as participation or drop-off.

```
# Step 1: Load and Inspect Dataset
print(" ++ Load and Inspect Dataset ++")
df = pd.read_excel("SLU Opportunity Wise Data-1710158595043.xlsx", engine="openpyxl")

print(df.head())
df.info()
df.isnull().sum()
```

3. Cleaning Steps Performed:

- All date-related columns were converted into proper datetime formats to enable time-based calculations and analysis.
- Missing values in categorical columns such as gender, country, and institution name were handled by assigning the value "Unknown" to maintain record completeness.
- Duplicate records were identified and removed to ensure data integrity.
- Column data types were reviewed and validated to avoid inconsistencies during feature engineering.

```python
# Step 2: Data Cleaning
print(" ++ Data Cleaning ++ ")
# Converting date columns properly :
date_cols = [
    'Learner SignUp DateTime',
    'Opportunity Start Date',
    'Opportunity End Date',
    'Apply Date',
    'Entry created at',
    'Date of Birth'
]
for col in date_cols:
    df[col] = pd.to_datetime(df[col], errors='coerce')

print("There Handling Missing Values ==>")

# for categorical columns:
categorical_cols = ['Gender', 'Country', 'Institution Name', 'Current/Intended Major']

df[categorical_cols] = df[categorical_cols].fillna("Unknown")
# Date of Birth missing values left as NaT (intentional)

print("There Removing Duplicates ==>")
df.drop_duplicates(inplace=True)
```

```
 ++ Data Cleaning ++
There Handling Missing Values ==>
     Learner SignUp DateTime  ... Opportunity Start Date
0        2023-06-14 12:30:35  ...    2022-03-11 18:30:39
1        2023-01-05 05:29:16  ...    2022-03-11 18:30:39
2        2023-09-04 20:35:08  ...    2022-03-11 18:30:39
3        2023-08-29 05:20:03  ...    2022-03-11 18:30:39
4        2023-06-01 15:26:36  ...    2022-03-11 18:30:39
...                       ...  ...                    ...
8553     2024-05-03 16:39:21  ...    2024-08-03 14:00:00
8554     2024-06-03 15:58:47  ...    2024-08-03 14:00:00
8555     2024-02-29 15:02:31  ...    2024-08-03 14:00:00
8556     2023-12-23 03:53:12  ...    2024-08-03 14:00:00
8557     2023-01-06 13:22:01  ...    2024-08-03 14:00:00

[8558 rows x 16 columns]
```

```
There Removing Duplicates ==>
     Learner SignUp DateTime  ... Opportunity Start Date
0        2023-06-14 12:30:35  ...    2022-03-11 18:30:39
1        2023-01-05 05:29:16  ...    2022-03-11 18:30:39
2        2023-09-04 20:35:08  ...    2022-03-11 18:30:39
3        2023-08-29 05:20:03  ...    2022-03-11 18:30:39
4        2023-06-01 15:26:36  ...    2022-03-11 18:30:39
...                       ...  ...                    ...
8553     2024-05-03 16:39:21  ...    2024-08-03 14:00:00
8554     2024-06-03 15:58:47  ...    2024-08-03 14:00:00
8555     2024-02-29 15:02:31  ...    2024-08-03 14:00:00
8556     2023-12-23 03:53:12  ...    2024-08-03 14:00:00
8557     2023-01-06 13:22:01  ...    2024-08-03 14:00:00

[8558 rows x 16 columns]
```

Issues Encountered and Resolution

One of the main challenges encountered was inconsistent formatting in date columns, which could lead to incorrect calculations. This issue was resolved by explicitly converting all relevant columns into datetime format and coercing invalid entries where necessary. Missing values in sensitive fields such as Date of Birth were intentionally left unfilled to avoid introducing incorrect assumptions.

4. Feature Engineering

Feature engineering was performed to enhance the dataset by creating new variables that provide additional analytical value.

New Features Created

- **Age:** Calculated using the learner's date of birth and signup date. This feature enables demographic analysis of learners.
- **Opportunity Duration (Days):** Derived by calculating the difference between opportunity start and end dates. This helps in understanding opportunity length.
- **Signup Month and Year:** Extracted from the learner signup date to support time-based trend analysis.
- **Days to Apply:** Calculated as the difference between signup date and application date, representing learner engagement speed.

```python
#  Step 3: Feature Engineering
print(" ++ Feature Engineering ++ ")
print("Calculating Age ==>")
df['Age'] = (
    (df['Learner SignUp DateTime'] - df['Date of Birth']).dt.days // 365
)
print(df)

print("Opportunity Duration ==>")
df['Opportunity_Duration_Days'] = (
    df['Opportunity End Date'] - df['Opportunity Start Date']
).dt.days

print(df)
print("Signup Month & Year ==>")
df['Signup_Month'] = df['Learner SignUp DateTime'].dt.month
df['Signup_Year'] = df['Learner SignUp DateTime'].dt.year

print(df)
print("Application Delay Feature ==>")
df['Days_To_Apply'] = (
    df['Apply Date'] - df['Learner SignUp DateTime']
).dt.days
print(df)
```

```
 ++ Feature Engineering ++
Calculating Age ==>
     Learner SignUp DateTime  ...   Age
0        2023-06-14 12:30:35  ...  21.0
1        2023-01-05 05:29:16  ...  22.0
2        2023-09-04 20:35:08  ...  21.0
3        2023-08-29 05:20:03  ...  24.0
4        2023-06-01 15:26:36  ...  23.0
```

```
Opportunity Duration ==>
     Learner SignUp DateTime  ... Opportunity_Duration_Days
0        2023-06-14 12:30:35  ...                     841.0
1        2023-01-05 05:29:16  ...                     841.0
2        2023-09-04 20:35:08  ...                     841.0
3        2023-08-29 05:20:03  ...                     841.0
4        2023-06-01 15:26:36  ...                     841.0
```

```
Signup Month & Year ==>
     Learner SignUp DateTime  ... Signup_Year
0        2023-06-14 12:30:35  ...      2023.0
1        2023-01-05 05:29:16  ...      2023.0
2        2023-09-04 20:35:08  ...      2023.0
3        2023-08-29 05:20:03  ...      2023.0
4        2023-06-01 15:26:36  ...      2023.0
```

```
Application Delay Feature ==>
     Learner SignUp DateTime  ... Days_To_Apply
0        2023-06-14 12:30:35  ...           0.0
1        2023-01-05 05:29:16  ...           0.0
2        2023-09-04 20:35:08  ...           NaN
3        2023-08-29 05:20:03  ...          12.0
4        2023-06-01 15:26:36  ...           0.0
```

Feature Engineering Examples

- Opportunity duration was created to analyze whether shorter or longer opportunities show different engagement patterns.
- Days to apply was engineered to identify how quickly learners apply after signing up, which can be useful for behavioral analysis.

All newly created features were added as separate columns without modifying the original data.

5. Data Validation:

After cleaning and feature engineering, several validation checks were performed to ensure data accuracy and consistency:

- Verified that no critical columns contained unexpected missing values.
- Reviewed summary statistics to confirm reasonable ranges for numerical features such as age and duration.
- Performed random sampling of records to manually validate calculated features.
- Ensured that derived features such as age and duration did not contain negative or illogical values.

```
 # Step4: Data Validation
print(" ++ Data Validation ++ ")
df.isnull().sum()
df.describe(include='all')
df.sample(5)

df = df[df['Age'] > 0]

print(df)
```

These checks confirmed that the dataset is consistent and suitable for further analysis.

```
 ++ Data Validation ++
     Learner SignUp DateTime  ... Days_To_Apply
0        2023-06-14 12:30:35  ...           0.0
1        2023-01-05 05:29:16  ...           0.0
2        2023-09-04 20:35:08  ...           NaN
3        2023-08-29 05:20:03  ...          12.0
4        2023-06-01 15:26:36  ...           0.0
```

6. Conclusion:

During Week 1, the dataset was successfully cleaned, standardized, and enhanced through feature engineering. Key issues such as missing values, duplicate records, and inconsistent date formats were resolved. Several meaningful features were created to improve the analytical depth of the dataset.

7. Next Steps:

In Week 2, the cleaned and engineered dataset will be used for exploratory data analysis and visualization. The focus will shift toward identifying trends, patterns, and insights related to learner behavior and opportunity performance.

8. Appendix

- Python scripts used for data cleaning and feature engineering
- Summary tables and sample records for validation
- Notes on assumptions and design decisions made during preprocessing